RSHS

MATH O MAGIC

1·3

by

Karen Anderson

EDUCATIONAL DIRECTIONS PUBLICATIONS

INTRODUCTION

MATH O MAGIC 1·3 is part of a series of 12 books

MATHOMAGIC 1·2	MATHOMAGIC 4·3
MATHOMAGIC 1·3	MATHOMAGIC 4·4
MATHOMAGIC 2·3	MATHOMAGIC 5·3
MATHOMAGIC 2·4	MATHOMAGIC 5·4
MATHOMAGIC 3·3	MATHOMAGIC 6·3
MATHOMAGIC 3·4	MATHOMAGIC 6·4

These books are designed for students in years eight to ten.

1·2 and 1·3 are designed to revise the skills from earlier years and to be a platform from which high school skills are developed.

2·3, 3·3, 4·3, 5·3 and 6·3 develop skills in the SPACE, MEASUREMENT, and INFERENCE strands.

2·4, 3·4, 4·4, 5·4 and 6·4 address skills in the NUMBER and FUNCTION strands.

Stages 2, 3, 4, 5 and 6 indicate the level of difficulty, with each following in a sequential manner.

PREFACE

Often students find the necessary skill practice in Mathematics to be very boring. This series of books has been designed as a motivational tool to help the students to feel more enthusiastic about their work.

These worksheets are designed to be used in various ways –

1. MOTIVATION – Students are keen to solve the puzzle and keep asking for more.

2. INCENTIVE – As a reward for students who finish their work early.

3. SELF CHECKING
   – For students to be able to check their own work as they go along.

4. EXTRA PRACTICE
   – A fun way of doing extra practice on skills development.

5. HOMEWORK – an interesting way to get students to do more consolidation of mathematical skills in a way they enjoy.

6. REVISION – An exciting way to stimulate revision.

However you use these sheets, you will always be asked for more. The students enjoy working out the puzzles and riddles and get a great sense of satisfaction from getting them right.

Copies of this book may be obtained from:

EDUCATIONAL DIRECTIONS PUBLICATIONS
11 CAVALIER COURT,
THORNLIE 6108,
WESTERN AUSTRALIA.
PHONE / FAX (09) 459 0546

COPYRIGHT © EDUCATIONAL DIRECTIONS PUBLICATIONS

First published 1993

Limited copying rights apply only to the original purchaser of this book, who may reproduce sufficient copies for non commercial classroom use.

Typeset and design artwork by Sharon Locke and David Locke

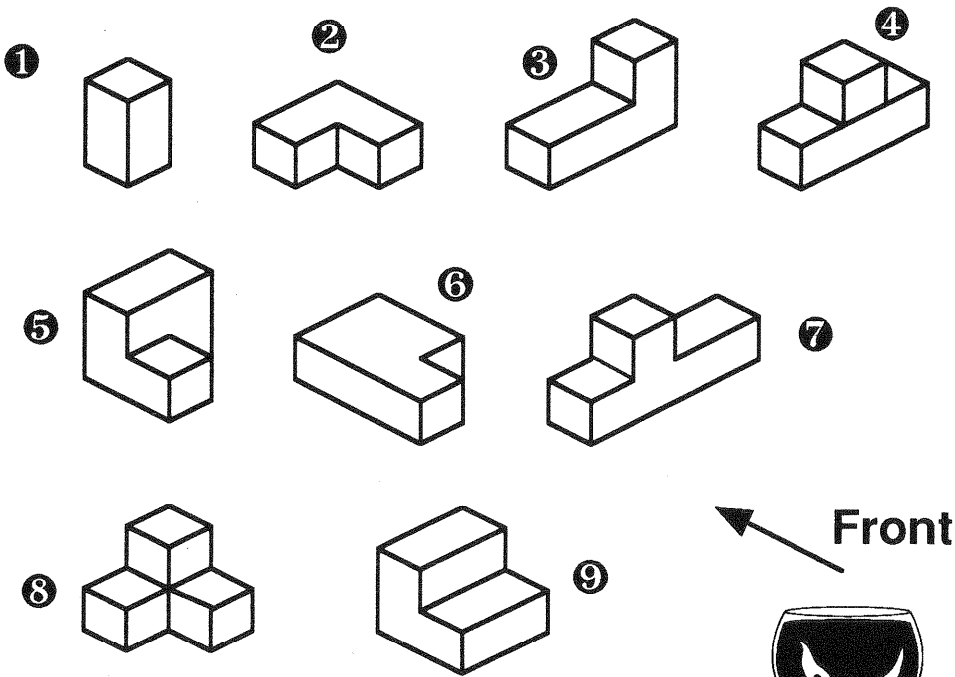
Publishing editor – Greg Hurn

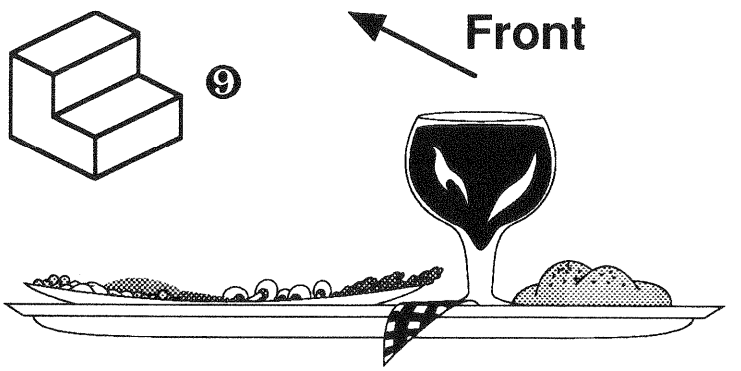
ISBN – 1 875712 00 3 (Set)
1 875712 12 7



WHO ALWAYS HAS GUESTS FOR DINNER?

Match each isometric sketch shown below with its orthogonal representation. Each pair has a letter and a number. Write the letter above the number in the code overleaf.



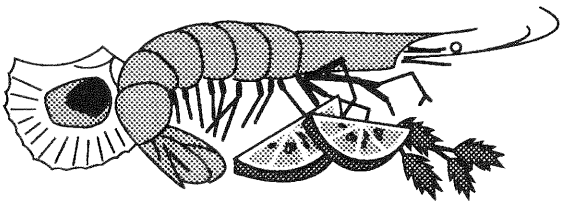


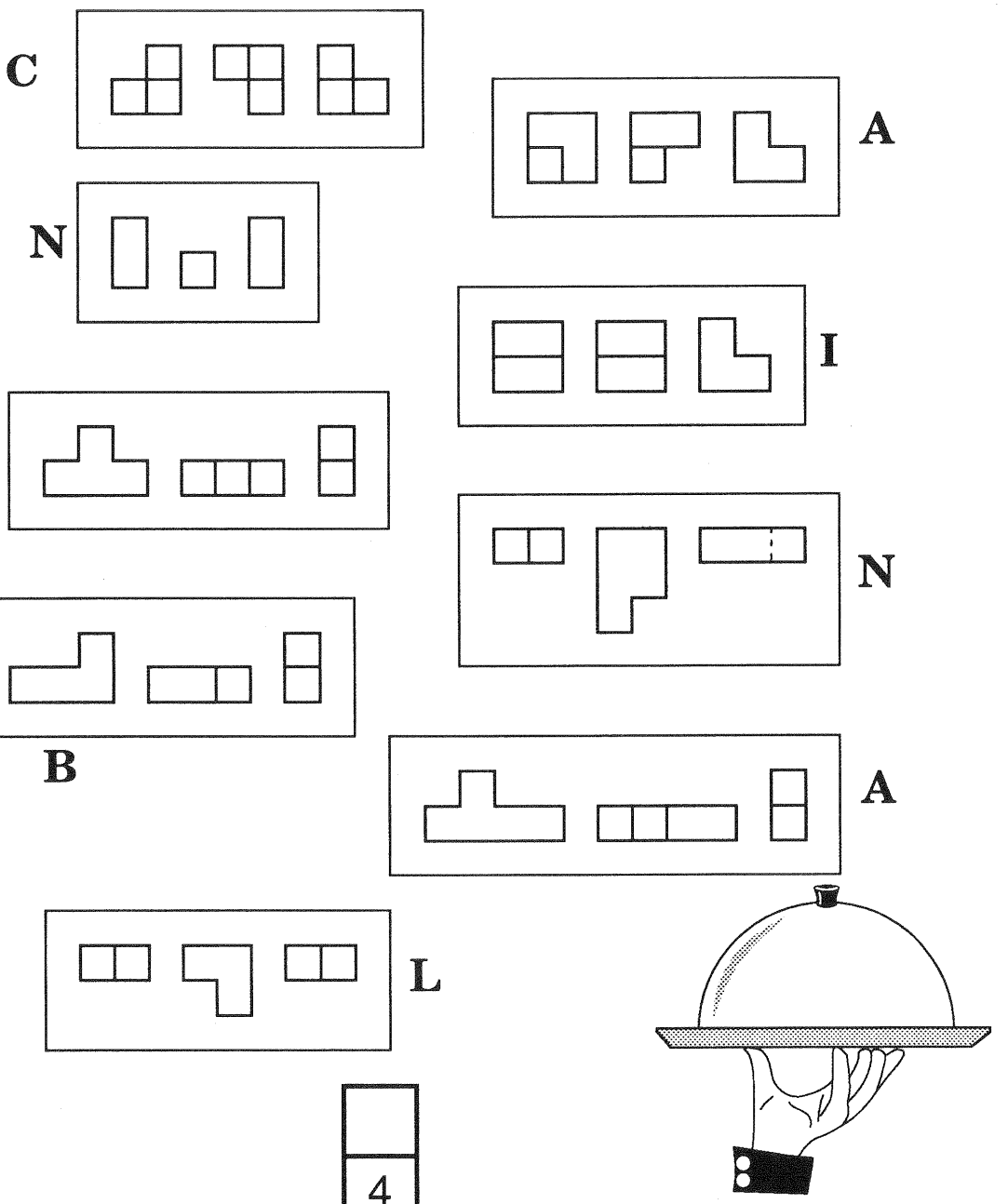
© EDUCATIONAL DIRECTIONS PUBLICATIONS

see next page

WHO ALWAYS HAS GUESTS FOR DINNER?

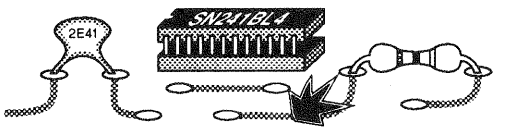
(continued)





© EDUCATIONAL DIRECTIONS PUBLICATIONS

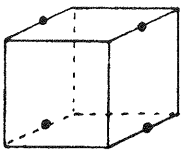
Do Robots Have Brothers?



INSTRUCTIONS: Imagine a cut was made in each solid model through the points shown. What shape cross – section would you have? Of the three choices, select the best one. Notice the letter next to it. Write that letter above the number of the question in the code.

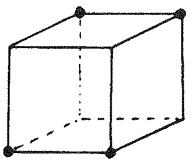
4 13 9 3 6 15 10 5 14 8 1 11 7 12 2

1



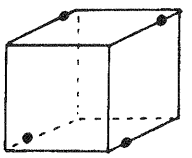
T S R

2



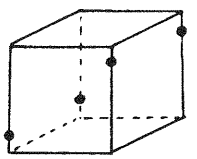
S D A

3



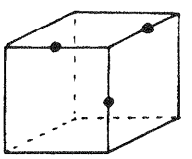
Y N W

4



D E O

5

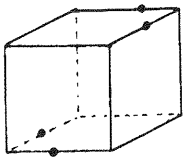


A N S

© EDUCATIONAL DIRECTIONS PUBLICATIONS

Do Robots Have Brothers?

6



T



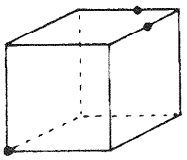
S



F



7



N



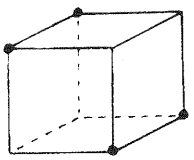
E



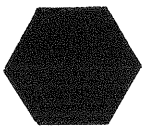
O



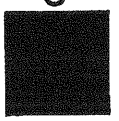
8



E



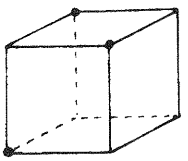
U



I



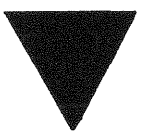
9



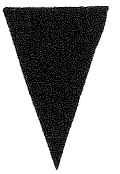
D



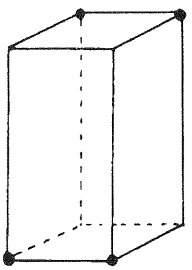
L



K



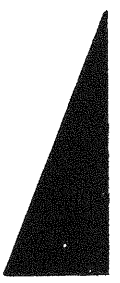
10



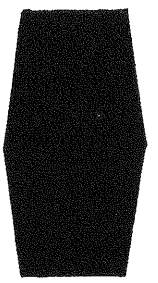
A



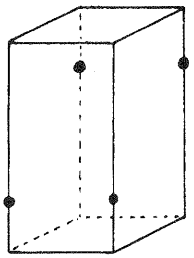
T



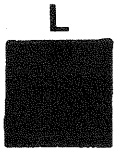
N



11



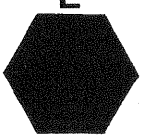
L



T



E



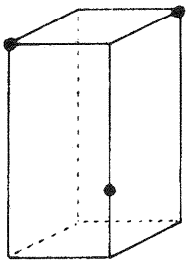
© EDUCATIONAL DIRECTIONS PUBLICATIONS

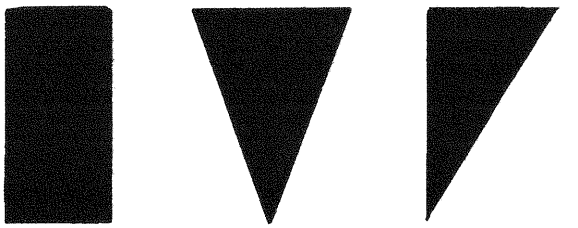



Do Robots Have Brothers?

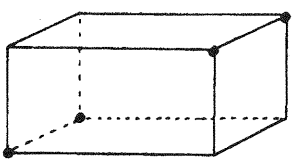


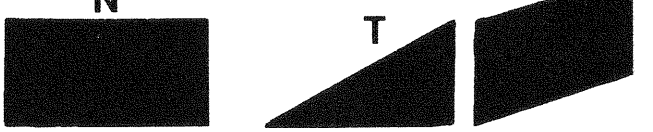

12



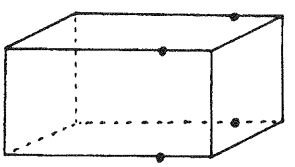


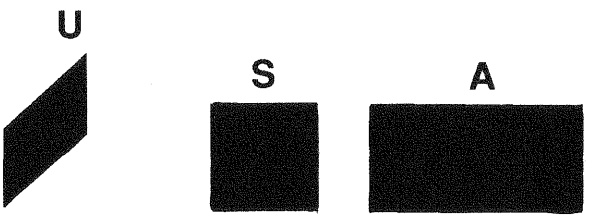
13



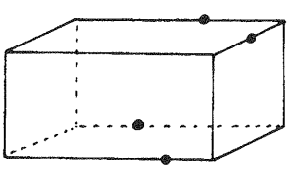


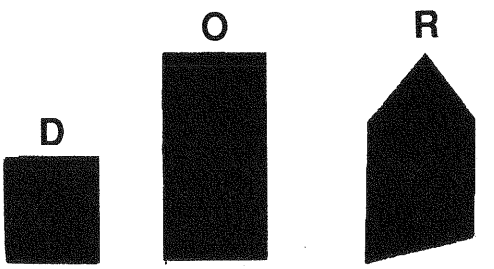
14





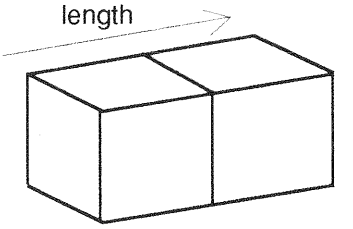
15





© EDUCATIONAL DIRECTIONS PUBLICATIONS

WHAT DOES A PIG USE TO WRITE WITH?



Use this model as the base model. It has a volume of 2 units and a surface area of 10 units.

For each of the following, circle the correct answer. Notice the letter next to it. Write that letter above the number of the question in the code below.

When the length only is doubled, the new

1.	volume is	8 – **S**	4 – **N**	2 – **T**
2.	surface area is	18 – **O**	8 – **R**	20 – **A**

When the height only is tripled, the new

3.	volume is	16 – **E**	8 – **L**	6 – **K**
4.	surface area is	6 – **A**	22 – **I**	30 – **S**

When the height and length only are doubled, the new

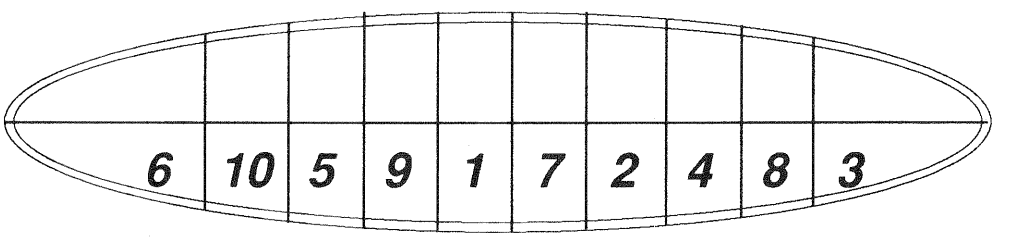
5.	volume is	8 – **N**	16 – **R**	12 – **D**
6.	surface area is	40 – **A**	22 – **E**	28 – **P**

When all dimensions are doubled, the new

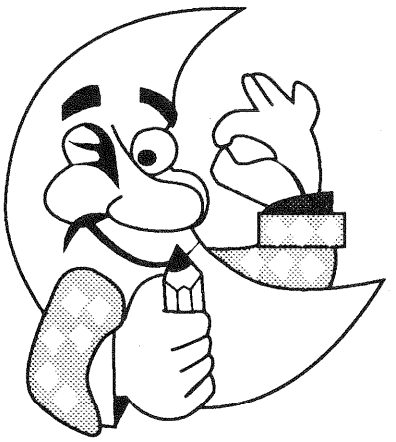
7.	volume is	4 – **R**	8 – **C**	16 – **D**
8.	surface area is	40 – **N**	20 – **L**	46 – **F**

When all dimensions are tripled, the new

9.	volume is	6 – **S**	54 – **A**	18 – **E**
10.	surface area is	54 – **I**	90 – **E**	72 – **T**

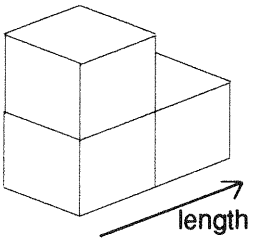


© EDUCATIONAL DIRECTIONS PUBLICATIONS



IF OXYGEN IS BREATHED IN THE DAYTIME, WHAT IS BREATHED IN THE NIGHT TIME?

Use this model as the base model. It has a volume of 3 units and a surface area of 14 units.



For each of the following, circle the correct answer. Notice the letter next to it. Write that letter above the number of the question in the code below.

When the height only is doubled, the new

1.	volume is	**9 – T**	**3 – D**	**6 – N**
2.	surface area is	**24 – O**	**28 – E**	**56 – D**

When the length only is doubled, the new

3.	volume is	**9 – F**	**6 – E**	**3 – A**
4.	surface area is	**24 – H**	**28 – L**	**56 – O**

When the length and height only are doubled, the new

5.	volume is	**12 – I**	**6 – D**	**24 – N**
6.	surface area is	**56 – E**	**28 – S**	**40 – R**

When all dimensions are doubled, the new

7.	volume is	**6 – A**	**12 – D**	**24 – N**
8.	surface area is	**40 – R**	**56 – T**	**70 – S**

When all dimensions are tripled, the new

9.	volume is	**9 – A**	**81 – G**	**27 – H**
10.	surface area is	**126 – G**	**42 – U**	**96 – N**

7	5	10	4	8	6	2	9	3	1

© EDUCATIONAL DIRECTIONS PUBLICATIONS

WHAT KIND OF TEETH CAN YOU BUY FOR A DOLLAR?

Decide whether the diagrams show planes of symmetry. If the diagram does show a plane of symmetry circle the number-letter pair in the YES column. If not, circle the number-letter pair in the NO column. Write the letter above the appropriate number in the code box.

	YES	NO
	9 — K	4 — M
	4 — B	1 — T
	9 — S	6 — H
	7 — E	6 — L
	3 — A	2 — U

	YES	NO
	8 — T	7 — U
	5 — O	3 — T
	2 — D	5 — C
	1 — E	8 — F

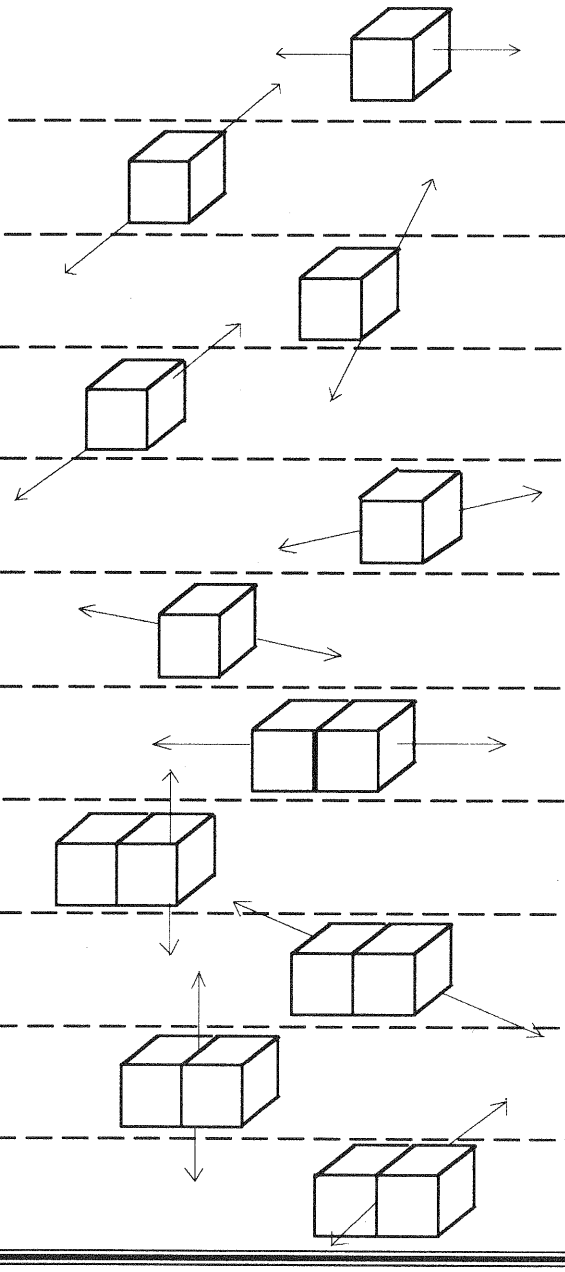
	4	2	5	9	3	7	1	8	6

© EDUCATIONAL DIRECTIONS PUBLICATIONS

WHAT KIND OF MAKE–UP DO GHOSTS USE?

4	2	1	11	9	6	11	1	3	7	8	10	2	5

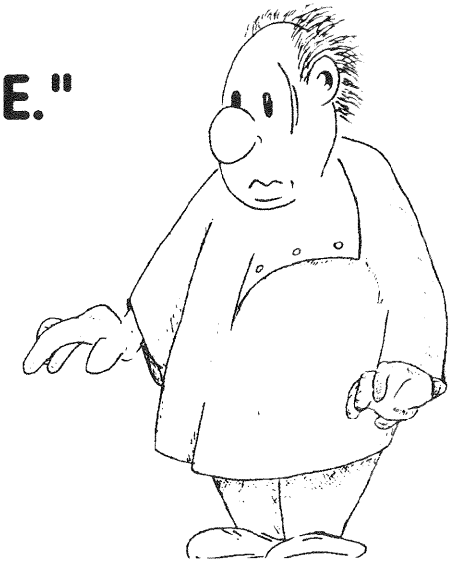
Decide whether the following diagrams of models made from cubes show axes of rotational symmetry. If the diagram does show an axis of symmetry, circle the number – letter pair in the YES column. If not, circle the number – letter pair in the NO column. Write the letter above the appropriate number in the code above.



YES	NO
3 – G	5 – D
8 – R	11 – Y
9 – T	6 – H
3 – O	4 – V
9 – S	7 – A
10 – F	1 – N
2 – A	8 – E
4 – T	5 – M
6 – N	7 – C
11 – I	1 – E
10 – E	2 – T

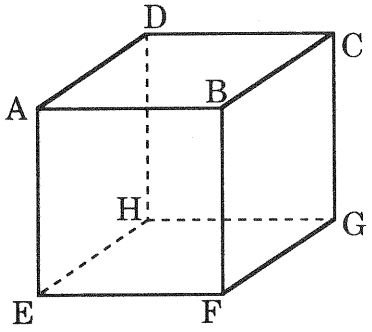
© EDUCATIONAL DIRECTIONS PUBLICATIONS

"DOCTOR! DOCTOR! PEOPLE KEEP IGNORING ME."



THE DOCTOR REPLIED . . .

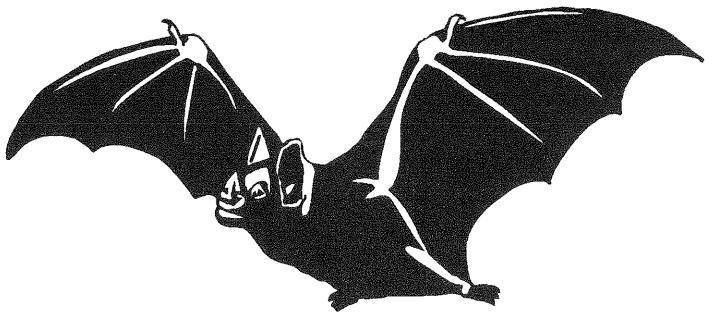
6	9	4	8	3	5	10	1	7	2



INSTRUCTIONS:
Examine the diagram of a cube shown to the left. Decide if each statement is true or false. Circle the number - letter pair in the appropriate column. Write the letter above the number in the code.

	TRUE	FALSE
✤ There are 8 vertices.	3 – P	4 – Y
✤ There are 8 faces.	6 – K	10 – E
✤ There are 8 edges.	1 – E	7 – S
✤ Face ABCD is parallel to face HGFE.	2 – E	3 – L
✤ Edge BF is horizontal.	9 – A	6 – N
✤ Edge EA is parallel to edge HD.	1 – A	2 – I
✤ Edges AB, CB and BF meet at one vertex.	4 – X	5 – M
✤ Face ABCD meets face FEAB at a single vertex.	10 – U	8 – T
✤ Face ABFE is congruent to face DCGH.	5 – L	7 – D
✤ The top face ABCD is shaped like a diamond.	8 – S	9 – E

© EDUCATIONAL DIRECTIONS PUBLICATIONS

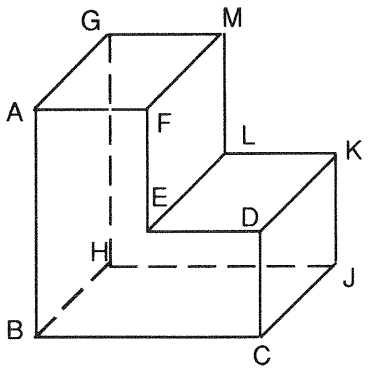


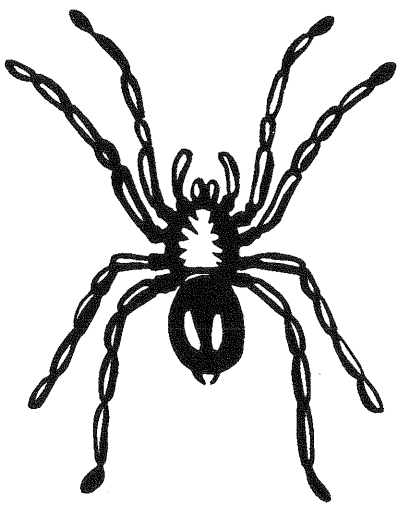
WHO DID THE VAMPIRE MARRY?

18	5	14	21	3	10	15	6	11	7	20	2

17	19	12	8	13	1	16	9	4

Shown below is a diagram of a model made from three cubes. Decide if each of the following 21 statements are true or false. Circle the number – letter pair in the appropriate column. Write the letter above the number in the code.





see next page

© EDUCATIONAL DIRECTIONS PUBLICATIONS

continued



WHO DID THE VAMPIRE MARRY?

	TRUE	FALSE
• Face ABCDEF is congruent to face GHJKLM.	3 – E	21 – T
• Face EDKL is diamond – shaped.	11 – F	16 – O
• Face FMLE is parallel to face DKJC.	5 – H	7 – I
• Face AFMG is congruent to face DKJC.	6 – S	17 – M
• Faces ELKD and KJCD intersect at vertex D.	19 – O	14 – E
• Faces FMLE and LKDE intersect at 2 points.	6 – T	4 – R
• Faces BHJC and DKJC are parallel.	20 – E	18 – T
• Face BHJC is twice the area of face FMLE.	9 – U	2 – S
• Faces ABHG and AGMF intersect at a single edge.	1 – B	12 – E
• The model has 13 vertices.	8 – J	15 – K
• The model has 8 faces.	10 – C	13 – M
• The model has 18 edges.	20 – O	18 – R
• Edges FM, EL, and CJ are all parallel.	8 – G	3 – U
• Vertex A is parallel to vertex D.	10 – D	13 – H
• Vertices A, F, M and G are all in the same plane.	19 – E	4 – D
• Vertices M, L, K and D are all in the same plane.	9 – E	21 – N
• Vertices G, M, K and J are in the same plane.	12 – I	16 – A
• Face AFEDCB is three times the area of face AFMG.	11 – D	5 – N
• Each face has the same number of vertices as it has edges.	17 – N	1 – D
• Face DKJC and edge LK do not intersect.	15 – L	7 – O
• Face AFEDCB and edge DK intersect at vertex D.	2 – R	14 – A



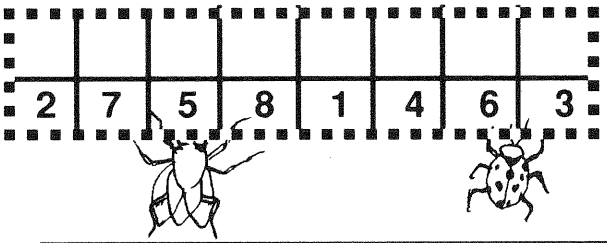


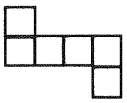
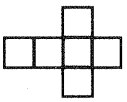
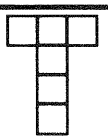
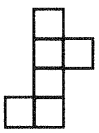
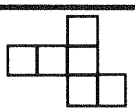
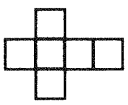
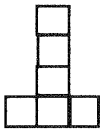


© EDUCATIONAL DIRECTIONS PUBLICATIONS

What Insects Were Found To Be Living On The Moon?

Shown below are some nets. Are they nets of **cubes** – yes or no? Circle the number – letter combination in the appropriate column. Write the letter above the number in the code. Keep working until you have the answer to the riddle.



	Net of a cube – YES	Net of a cube – NO
	6 – C	5 – E
	7 – A	3 – S
	1 – T	2 – S
	8 – A	1 – R
	2 – L	6 – M
	4 – I	8 – N
	5 – N	3 – F
	4 – E	7 – U





© EDUCATIONAL DIRECTIONS PUBLICATIONS

WHAT DO YOU GET WHEN YOU CROSS A LAWN MOWER WITH A BUDGIE?

SHOWN BELOW
ARE THREE VIEWS OF THE SAME CUBE.
IN THE CODE BELOW, WRITE EACH LETTER ANSWER
ABOVE THE NUMBER OF THE QUESTION.

4	8	13	7	3	11	10	5	9	1	6	12	2

1. Which face is opposite to X?



2. Which face is opposite to O?



3. Which face is opposite to E?



4. Which face is opposite to F?



5. Which face is opposite to Y?



6. Which face is opposite to L?

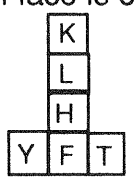


7. Which face is opposite to T?

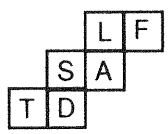


HERE ARE SOME NETS OF CUBES. WRITE THE LETTER ANSWER ABOVE THE NUMBER OF THE QUESTION IN THE CODE ABOVE.

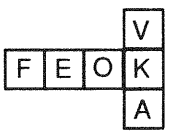
8. Which face is opposite to K?



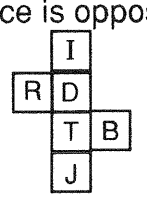
9. Which face is opposite to A?



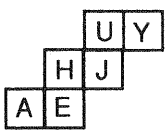
10. Which face is opposite to K?

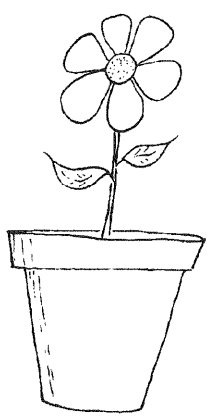


11. Which face is opposite to J?

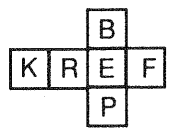


12. Which face is opposite to U?

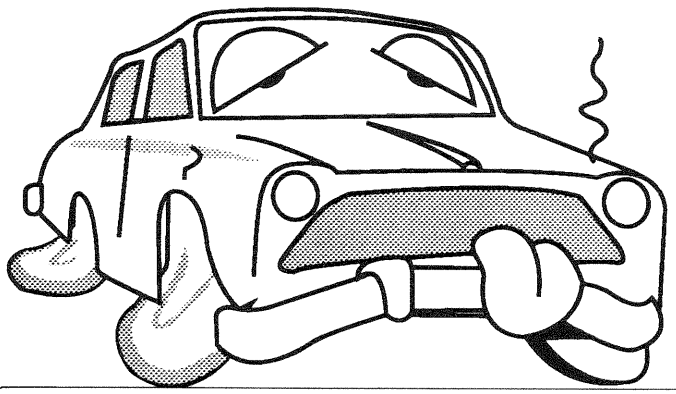




13. Which face is opposite to F?

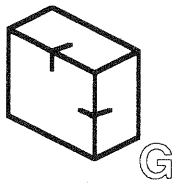


© EDUCATIONAL DIRECTIONS PUBLICATIONS

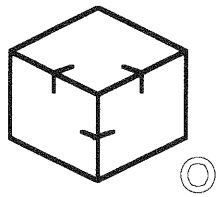


WHAT GOES PUTT . . . PUTT . . . PUTT . . . PUTT . . .?

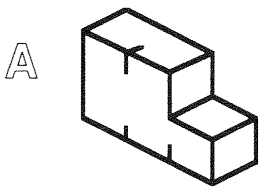
Shown below are some models made of cubes. The length of each cube is 1 cm. Determine the **surface area** of each model. Find your answer in the code below. Write the letter of each model above its surface area.

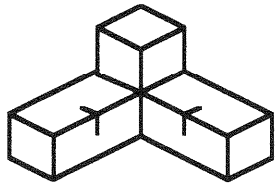

G

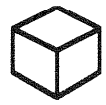

R

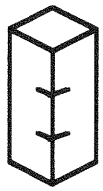

O

E

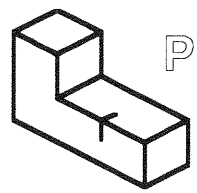

A




L


F




P

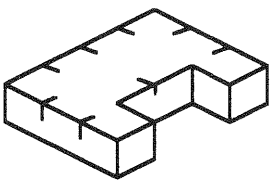
					20 cm^2
18 cm^2	24 cm^2	24 cm^2	10 cm^2		
16 cm^2	24 cm^2	6 cm^2	14 cm^2	26 cm^2	10 cm^2

© EDUCATIONAL DIRECTIONS PUBLICATIONS

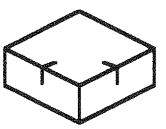
WHAT IS A TWIP?

Shown below are some models made from cubes. The length of each cube is 1 cm. Determine the surface area of each model. Look for your answer in the code below. Write the letter of each model above its surface area.

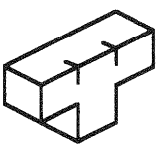
E

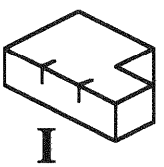


T



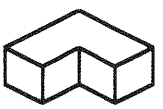
N



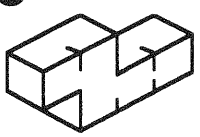


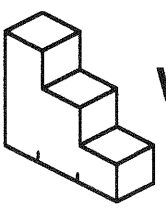
I

A

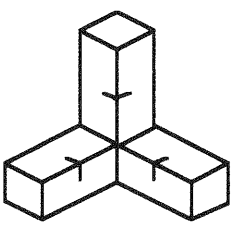


O





W



D

14 cm^2	24 cm^2	20 cm^2	30 cm^2	36 cm^2	22 cm^2	18 cm^2

14 cm^2	16 cm^2	24 cm^2	14 cm^2	20 cm^2	18 cm^2

© EDUCATIONAL DIRECTIONS PUBLICATIONS

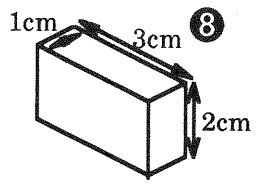
WHAT IS ANOTHER NAME FOR A TELEPHONE BOOTH ?

4	3	2	4	6	6	7	5	9	8	1

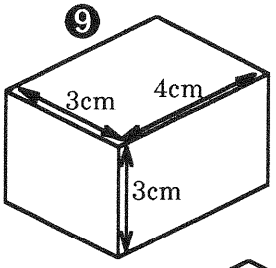
Determine the ***surface area*** of the following. Look for your answers in the Code Key. Notice the letter next to each answer. Write that letter above the number of the question in the code.

CODE KEY

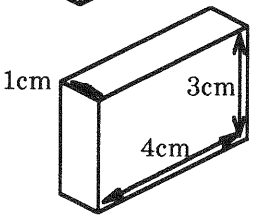
- 34 cm^2 – R
- 38 cm^2 – X
- 42 cm^2 – C
- 22 cm^2 – O
- 32 cm^2 – H
- 28 cm^2 – T
- 52 cm^2 – A
- 66 cm^2 – B
- 76 cm^2 – E



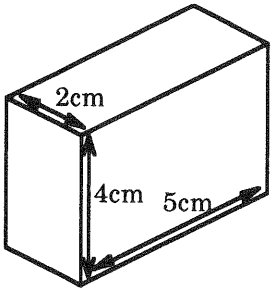
8



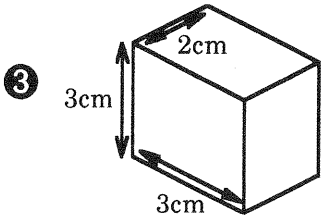
9



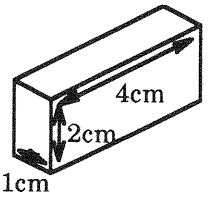
1



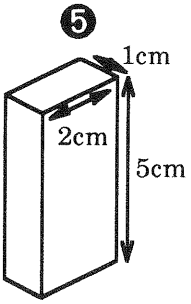
7



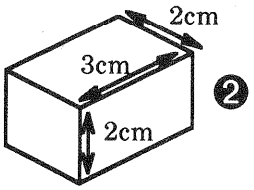
3



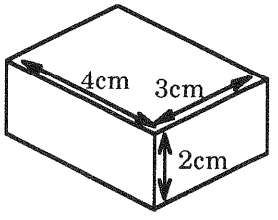
6



5



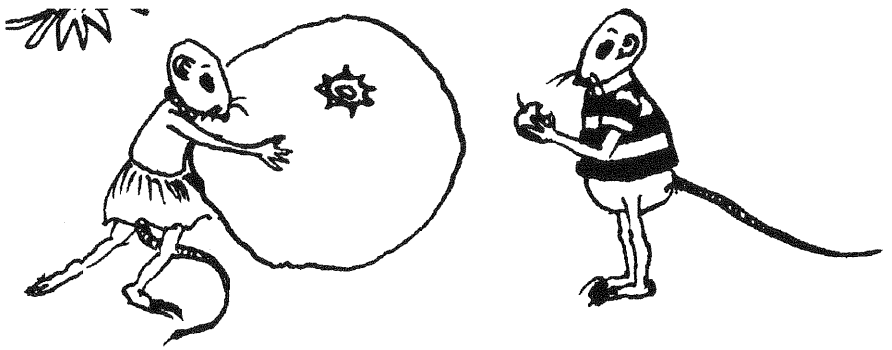
2



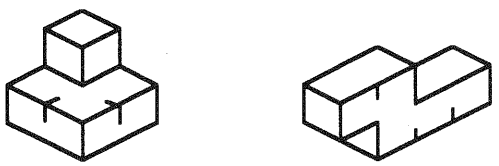
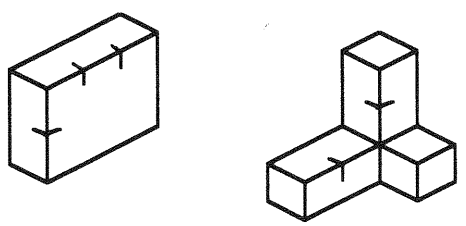
4

© EDUCATIONAL DIRECTIONS PUBLICATIONS

WHAT WOULD YOU GET IF YOU CROSSED DRACULA WITH A PIXIE?



Shown below are 9 pairs of models. Each pair has the same volume. Determine whether the surface area of each pair is the same or different. Circle the number – letter pair in the appropriate column. Write the letter above the number in the code overleaf.

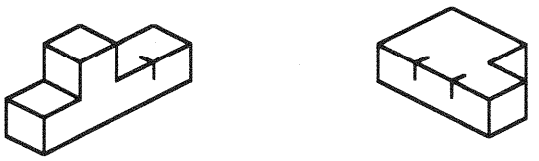
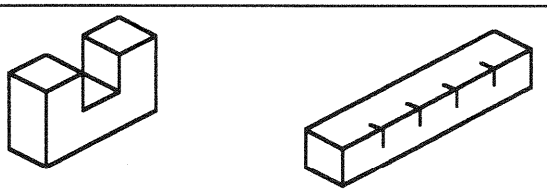
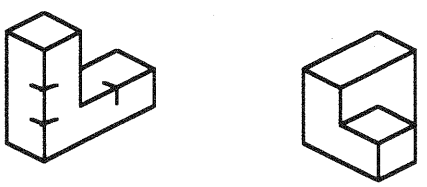
	Same surface area	Different surface area
	4 – G	2 – I
	8 – E	6 – A
	3 – Y	1 – A

see next page

© EDUCATIONAL DIRECTIONS PUBLICATIONS

continued

WHAT WOULD YOU GET IF YOU CROSSED DRACULA WITH A PIXIE?

	Same surface area	Different surface area
	7 – S	4 – O
	1 – R	5 – M
	9 – N	8 – H
	5 – F	6 – U
	2 – M	9 – E
	7 – T	3 – F



7 8 9 7 4 4 7 8 5 6 2 1 3

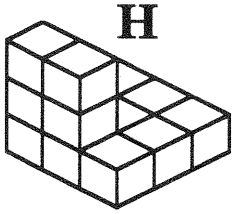
© EDUCATIONAL DIRECTIONS PUBLICATIONS



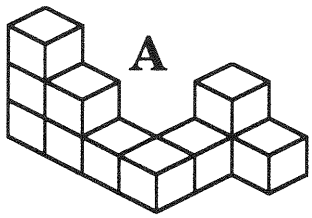
Determine the number of cubic units making up the volume of each of the following. Look for your answers in the code below. Write the letter of each model above its volume.

WHY IS A BRIDE ALWAYS OUT OF LUCK ON HER WEDDING DAY?

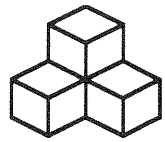
H



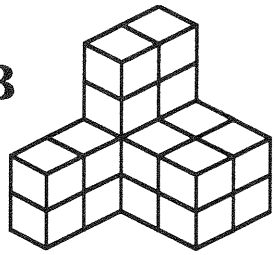
A



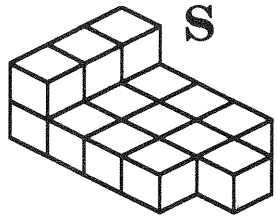
M



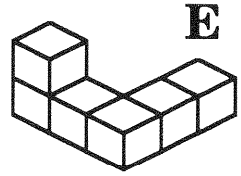
B



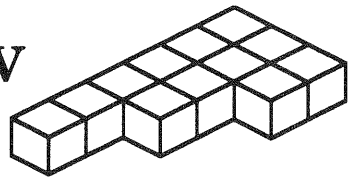
S



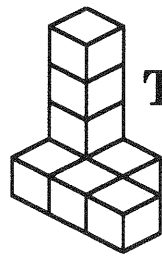
E



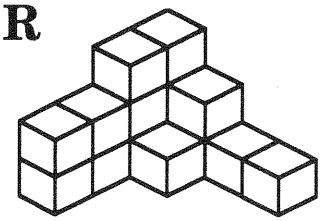
V



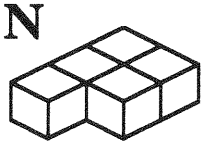
T



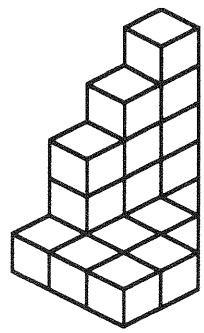
R



N



I



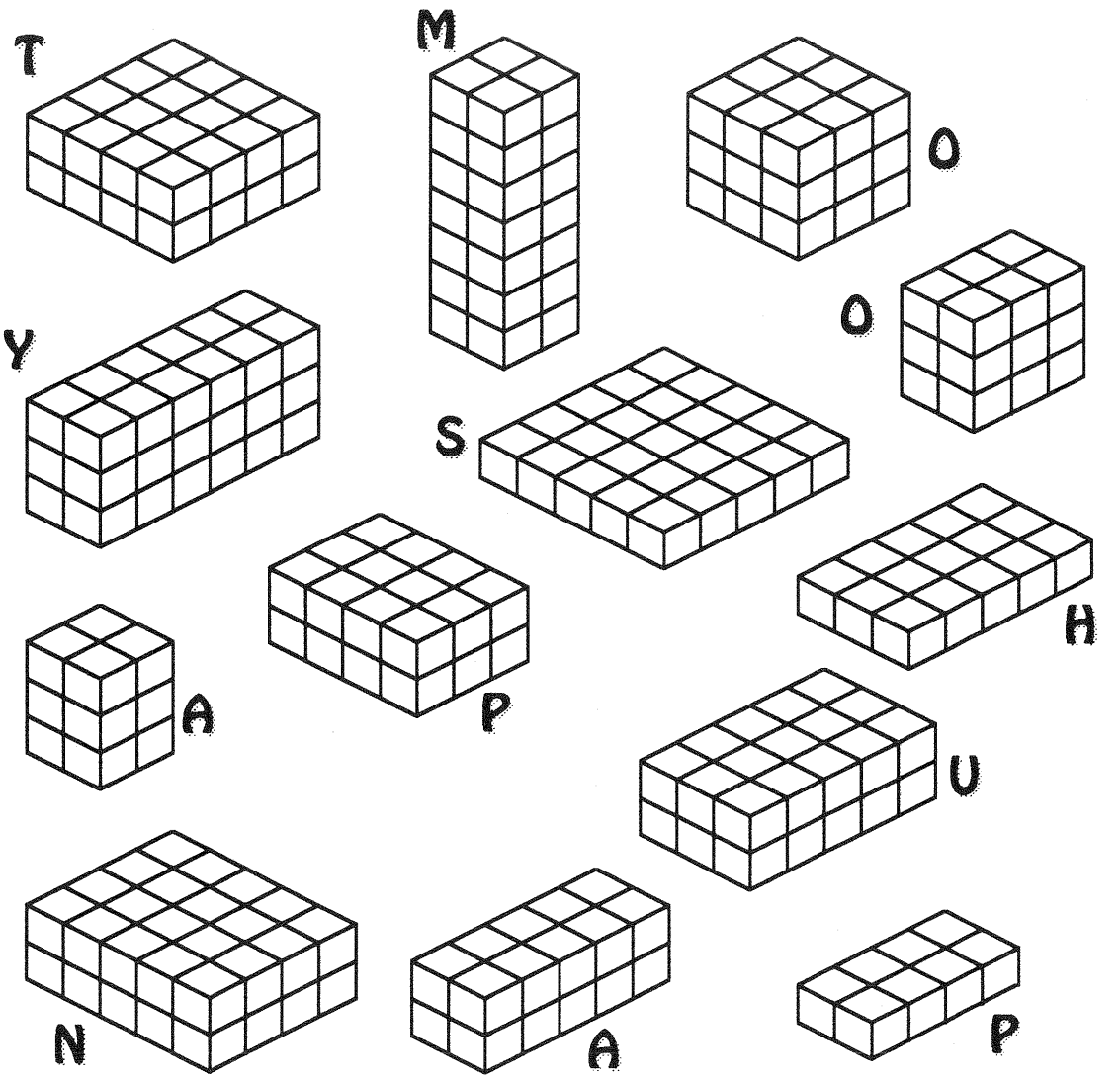
16	13	6	5	6	12	6	15	4	11	15	15	18	6	16

8	13	6	20	6	16	8	4	11	5

© EDUCATIONAL DIRECTIONS PUBLICATIONS

WHAT DO YOU CALL A LARGE ANIMAL THAT CAN PUT OTHERS TO SLEEP?

Make each of the models below with centicubes. Find the volume of each model by counting the number of cubic units. Look for your answers in the code below. Write the letter of each model above its volume.

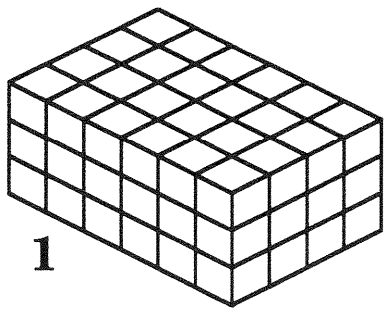


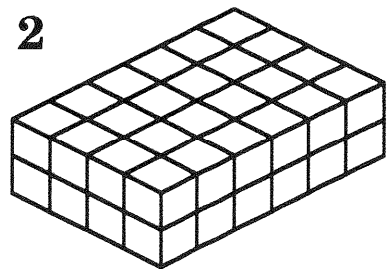
20 15 36 8 40 18 24 27 32 12 28 30 25

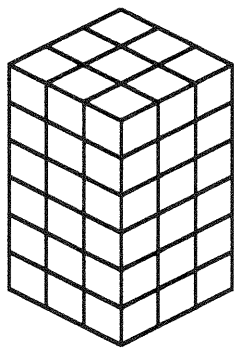
© EDUCATIONAL DIRECTIONS PUBLICATIONS

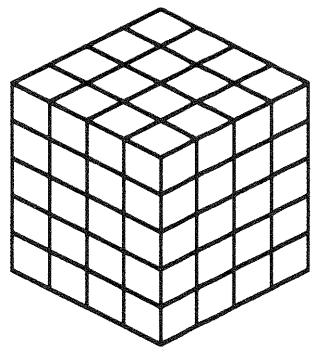
What Would You Get If You Crossed A Giraffe With A Porcupine?

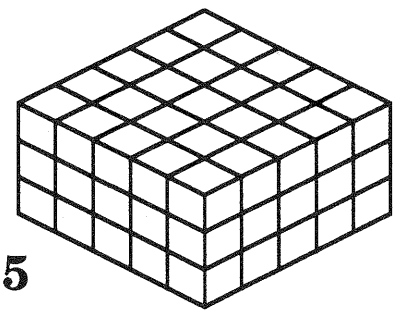
Make each of the following models with centicubes. Determine the volume of each model. Look for your answers in the Code Key. Notice the letter next to each answer. Write that letter above the number of each question in the puzzle. Keep working until you have the answer to the riddle.

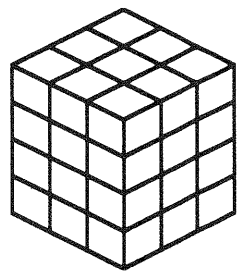

1

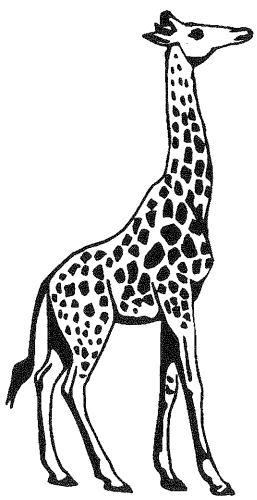

2

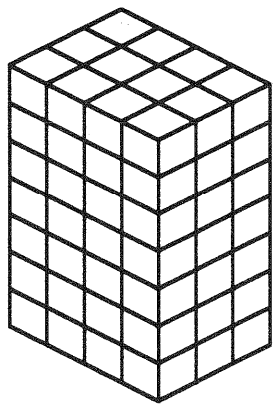

3

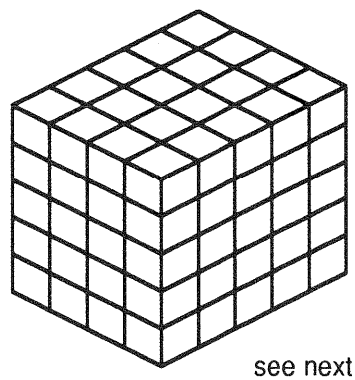

4


5


6




7

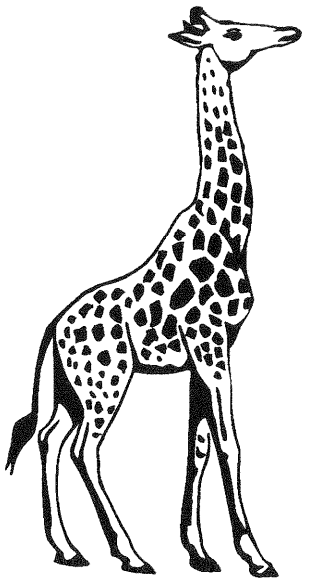

8

see next page

© EDUCATIONAL DIRECTIONS PUBLICATIONS

continued

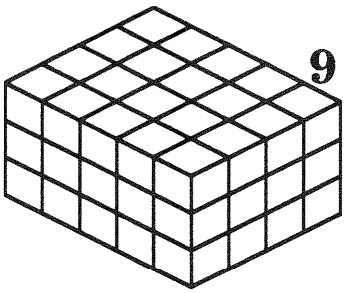
What Would You Get If You Crossed A Giraffe With A Porcupine?

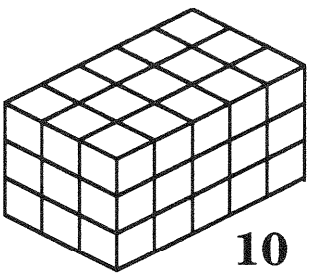


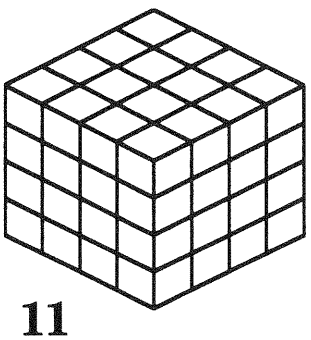
3		4	12	8		9	6	4	10	6

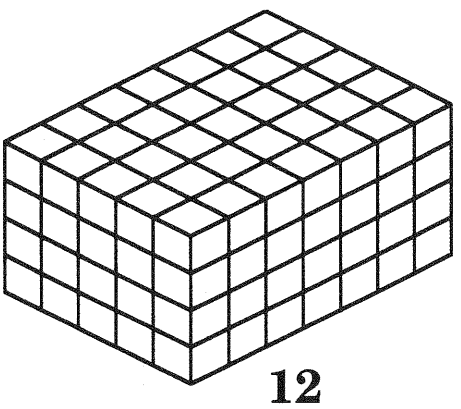
			1	8	2	14				

4	8	8	4	11	5	10	13	7	11	



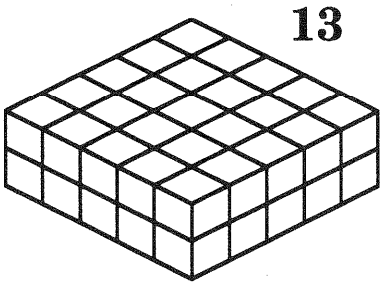


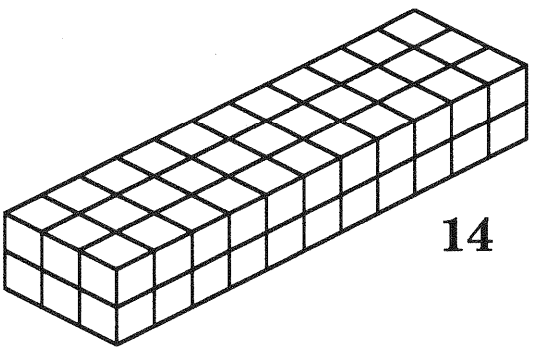




CODE KEY

80 cm^3 – T
45 cm^3 – R
75 cm^3 – B
36 cm^3 – E
66 cm^3 – G
100 cm^3 – O
140 cm^3 – W
84 cm^3 – S
48 cm^3 – N
50 cm^3 – U
60 cm^3 – M
64 cm^3 – H
72 cm^3 – L
54 cm^3 – A



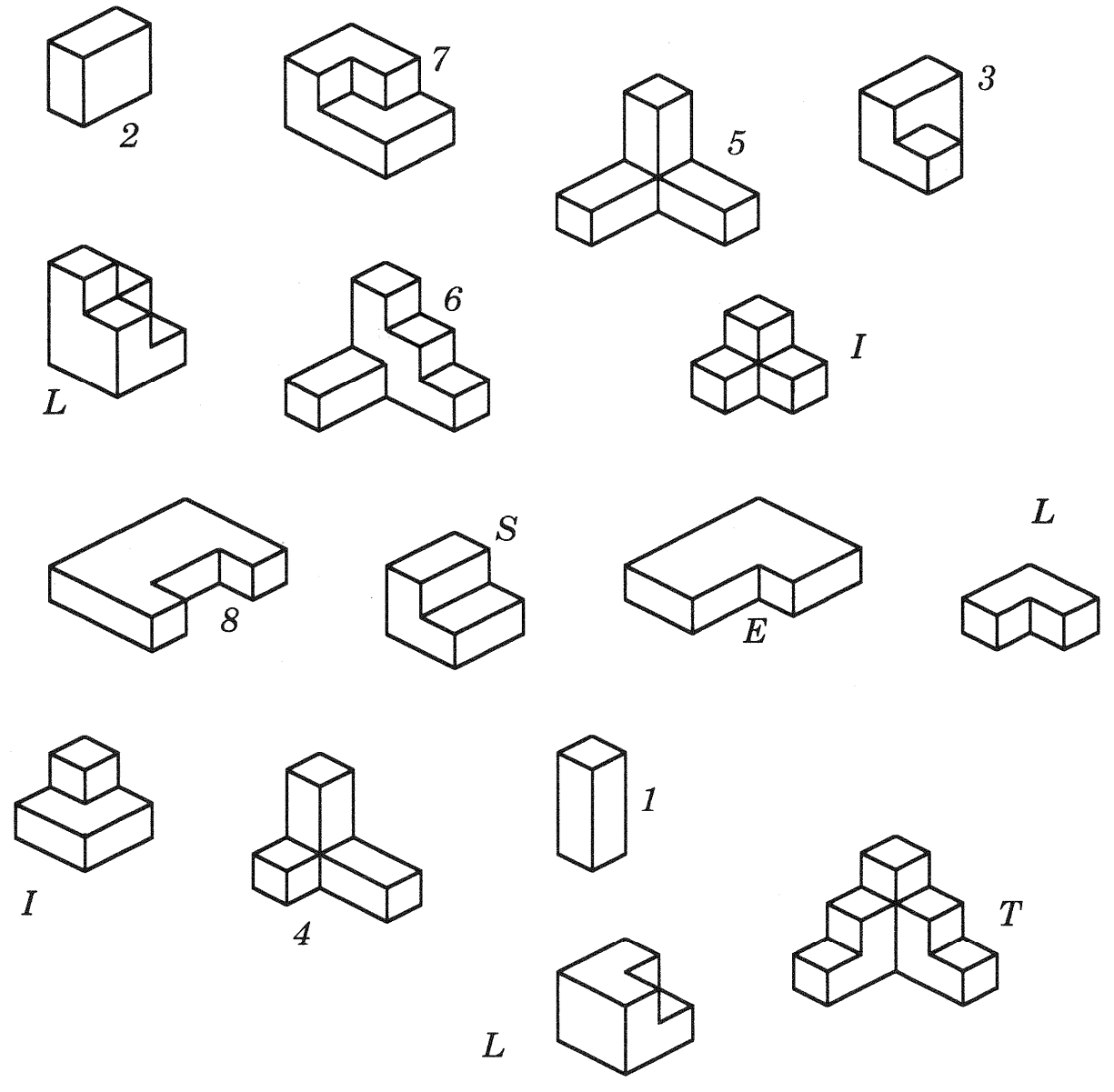


© EDUCATIONAL DIRECTIONS PUBLICATIONS

WHAT DO LIARS DO AFTER THEY DIE?



FROM THE MODELS SHOWN BELOW, FIND PAIRS HAVING THE SAME **VOLUMES**. WRITE THE LETTER OF EACH PAIR ABOVE THE MATCHING NUMBER IN THE CODE ABOVE.



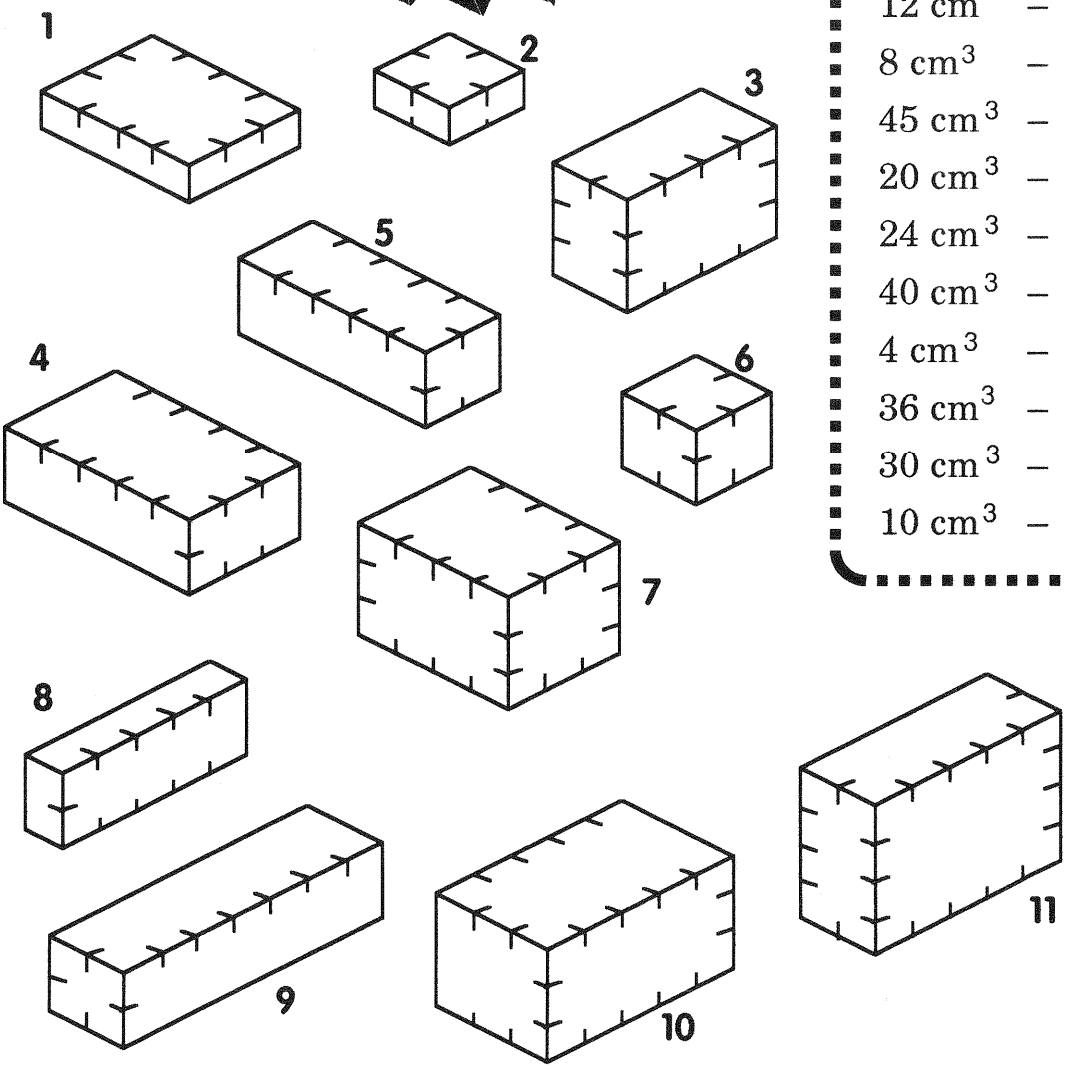
© EDUCATIONAL DIRECTIONS PUBLICATIONS

WHAT DID THE CANNIBAL SAY WHEN HE FOUND THE HUNTER ASLEEP?

Determine the volume of the following. Each model is made from cubes 1 cm in length. Look for your answers in the Code Key. Notice the letter next to each answer. Write that letter above the number of the question in the puzzle.

CODE KEY

28 cm^3	–	**E**
12 cm^3	–	**R**
8 cm^3	–	**A**
45 cm^3	–	**S**
20 cm^3	–	**F**
24 cm^3	–	**T**
40 cm^3	–	**B**
4 cm^3	–	**I**
36 cm^3	–	**N**
30 cm^3	–	**D**
10 cm^3	–	**K**



11	*1*	*9*	*6*	*8*	*5*	*6*	*10*	*3*	*2*	*7*	*11*	*9*	*4*

© EDUCATIONAL DIRECTIONS PUBLICATIONS

WHY DID THE BUTCHER PUT BELLS ON HIS SCALES?

Complete each of the following conversions. Look for your answers in the Code Key. Notice the letter next to each answer. Write that letter above the number of the question in the puzzle.

1. 4 L = _____ mL
2. 2·8 L = _____ mL
3. 0·75 L = _____ mL
4. 4 200 mL = _____ L
5. 9 800 L = _____ kL
6. 9 800 L = _____ mL
7. 26·5 L = _____ kL
8. 600 mL = _____ L
9. 45 mL = _____ L
10. 3·5 kL = _____ mL
11. 75 000 mL = _____ L
12. 0·85 kL = _____ L
13. 3 500 mL = _____ L
14. 1·75 kL = _____ L
15. 75 000 mL = _____ kL

CODE KEY

3·5	**O**
9·8	**W**
4 000	**J**
3 500 000	**E**
0·075	**S**
750	**A**
0·6	**L**
2 800	**D**
75	**C**
9 800 000	**T**
1 750	**G**
0·045	**U**
850	**I**
0·0265	**N**
4·2	**H**

15 13 4 10 11 13 9 8 2 1 12 7 14 8 10

3 8 8 6 4 10 5 10 12 14 4

© EDUCATIONAL DIRECTIONS PUBLICATIONS

WHY IS AN OPERATION FUNNY?



INSTRUCTIONS:
Complete each of the following conversions. Look for your answer in the grid below. Cross out each box containing a correct answer. When you finish there will be 7 boxes left. Read the remaining words in order from left to right.



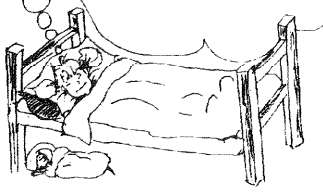
1. 7·5 L = _____ mL
2. 0·62 L = _____ mL
3. 950 mL = _____ L
4. 3·7 kL = _____ L
5. 0·58 kL = _____ L
6. 0·9 kL = _____ mL
7. 80 mL = _____ L
8. 0·07 L = _____ mL
9. 14 200 L = _____ kL
10. 450 L = _____ kL
11. 230 mL = _____ L
12. 12·3 L = _____ mL
13. 65 mL = _____ L

3 700	*0·23*	*0·7*	*0·95*
CAN	*DOCTOR*	*BECAUSE*	*HOSPITAL*
1 470	*12 300*	*580*	*9 000*
IT	*JOKE*	*TOLD*	*LEAVES*
0·08	*7 500*	*8*	*0·45*
BANDAGE	*WILL*	*THE*	*NURSE*
0·065	*14·2*	*900 000*	*62*
LAUGH	*NOT*	*DO*	*PATIENT*
7	*70*	*95*	*620*
IN	*FUNNY BONE*	*STITCHES*	*INJECTION*

© EDUCATIONAL DIRECTIONS PUBLICATIONS

WHY DID THE WORKER GET FIRED FROM HER JOB IN THE MATTRESS FACTORY?

Complete each of the following conversions. Look for your answers in the Code Key. Notice the letter next to each answer. Write that letter above the number of the question in the puzzle below.



1. $2\ m^3 =$ _____ cm^3
2. $1{\cdot}2\ m^3 =$ _____ cm^3
3. $700\ cm^3 =$ _____ m^3
4. $2\ 400\ cm^3 =$ _____ m^3
5. $9\ 500\ 000\ cm^3 =$ _____ m^3
6. $0{\cdot}7\ m^3 =$ _____ cm^3
7. $120\ 000\ cm^3 =$ _____ m^3
8. $2\ 400\ 000\ cm^3 =$ _____ m^3
9. $0{\cdot}95\ m^3 =$ _____ cm^3
10. $0{\cdot}2\ m^3 =$ _____ cm^3
11. $67\ 000\ cm^3 =$ _____ m^3
12. $4{\cdot}5\ m^3 =$ _____ cm^3
13. $6\ 800\ 000\ cm^3 =$ _____ m^3
14. $0{\cdot}45\ m^3 =$ _____ cm^3
15. $6{\cdot}7\ m^3 =$ _____ cm^3
16. $0{\cdot}5\ m^3 =$ _____ cm^3
17. $800\ 000\ cm^3 =$ _____ m^3

CODE KEY

0·0024	– S
2 000 000	– N
4 500 000	– W
6·8	– O
0·8	– H
950 000	– L
0·067	– C
1 200 000	– T
700 000	– I
6 700 000	– U
2·4	– D
500 000	– E
9·5	– G
450 000	– J
200 000	– B
0·0007	– Y
0·12	– A

4	17	16	12	7	4	11	7	15	5	17	2

9	3	6	1	5	8	13	12	1

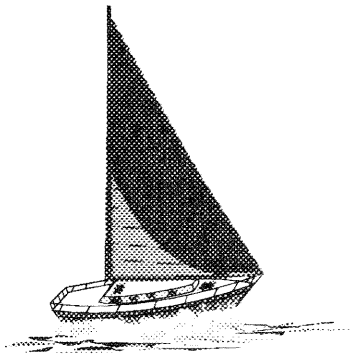
13	1	2	17	16	14	13	10



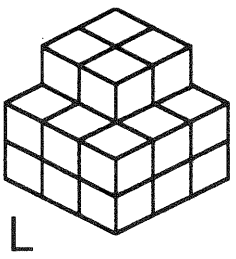
© EDUCATIONAL DIRECTIONS PUBLICATIONS

WHO SAILED WITH THE GHOST SHIP?

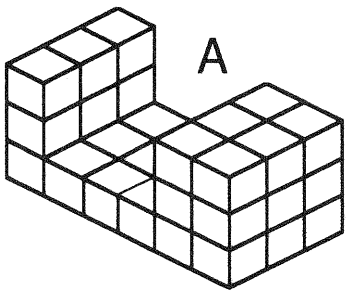
36	14	40	20	22	20	41	16	45	24	25	20	19



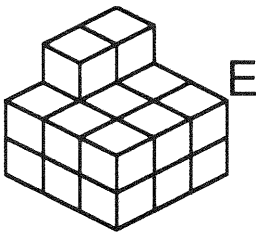
Each of the following models have been made with centicubes. Determine the volume of each model. Look for the volume in the code above. Write the letter of each model above its volume.



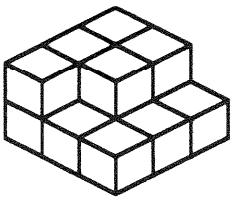
L



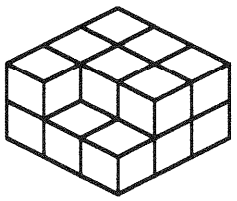
A



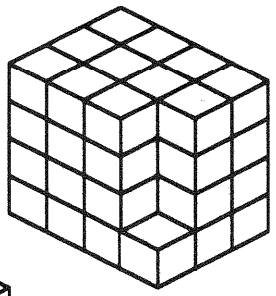
E



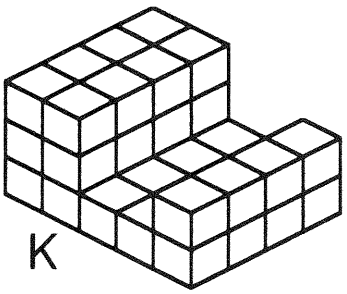
S



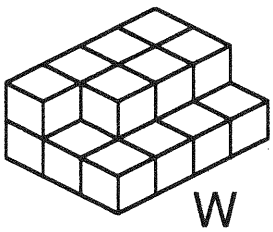
O



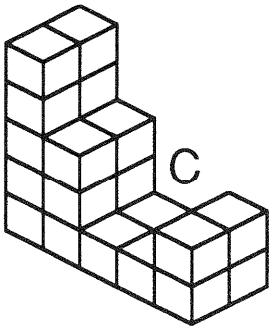
N



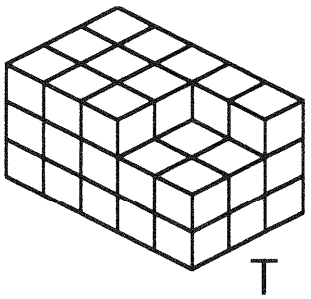
K



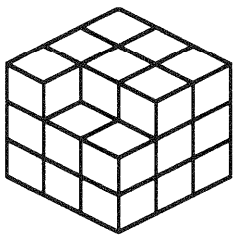
W



C



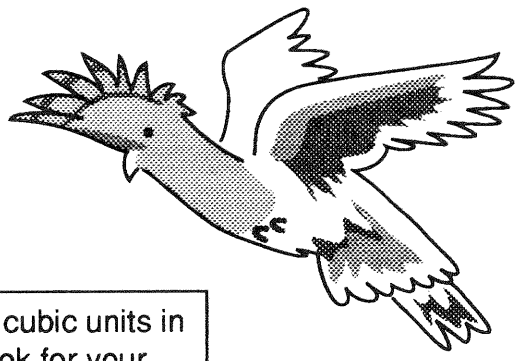
T

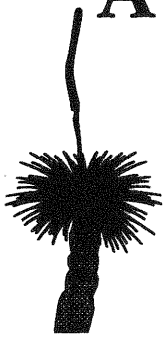


R

© EDUCATIONAL DIRECTIONS PUBLICATIONS

WHAT IS THE DEFINITION OF A PARROT?

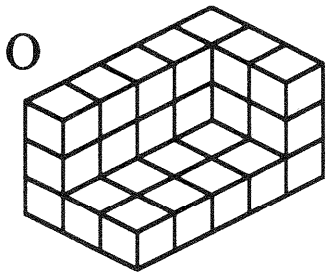


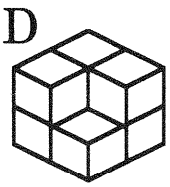


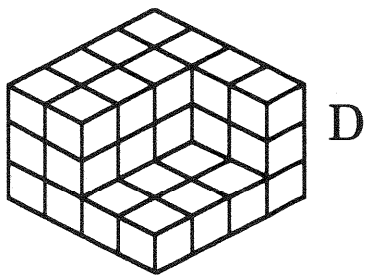
Determine the number of cubic units in each of these models. Look for your answer in the code below. Write the letter of each model above its answer.

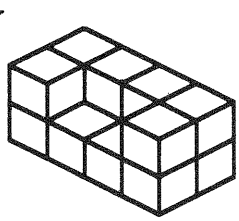
A ___ ___ ___ ___ ___ ___ ___ ___ ___ ___

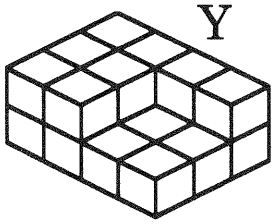
15 29 26 36 14 23 35 11 7 20

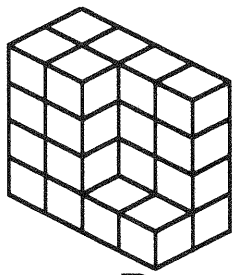


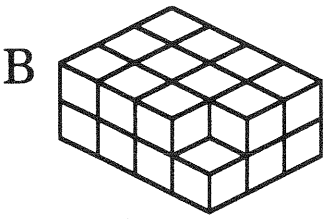


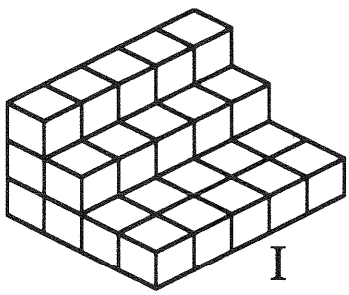


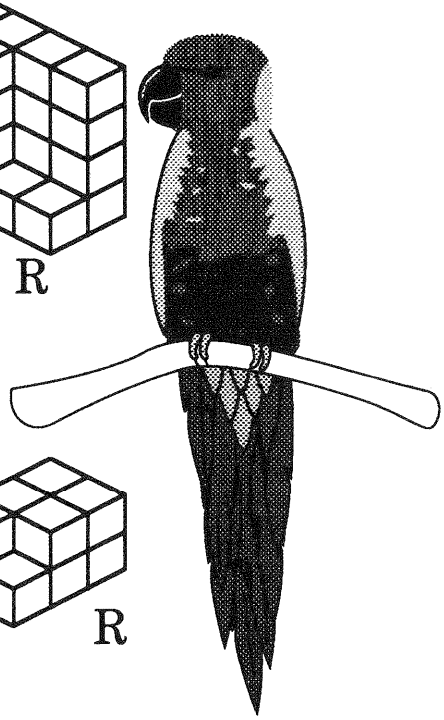




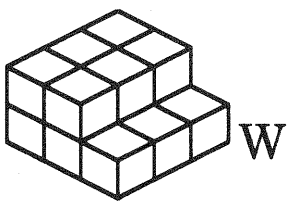


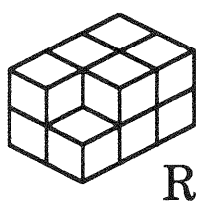












© EDUCATIONAL DIRECTIONS PUBLICATIONS

WHAT DID THE FIREFIGHTER'S WIFE FIND ON CHRISTMAS MORNING?

4	14	4	10	10	3	12	9	8		
5	3	12	2	7	6	13	1	9	8	11

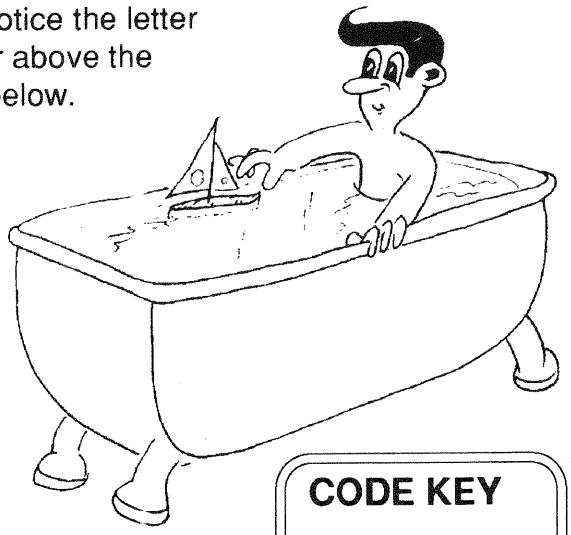
From the 3 choices given below, decide which unit would best be used to measure each item. Circle your choice. Notice the letter next to it. Write that letter above the number of the question in the code above.

1.	the mass of an apple	**N**: kg	**K**: g	**R**: mg
2.	the distance from Sydney to Melbourne	**S**: km	**T**: m	**D**: cm
3.	the mass of a tractor	**N**: g	**A**: kg	**E**: t
4.	the capacity of a cup	**A**: mL	**O**: L	**D**: kL
5.	the thickness of a piece of cardboard	**R**: m	**A**: cm	**H**: mm
6.	the mass of a tablet	**S**: kg	**E**: g	**O**: mg
7.	the capacity of a reservoir	**R**: L	**T**: kL	**F**: mL
8.	the volume of an orange	**N**: cm^3	**G**: m^3	**A**: mL
9.	the volume of a refrigerator	**I**: m^3	**T**: cm^3	**A**: L
10.	the mass of your maths file	**D**: g	**E**: mg	**P**: kg
11.	the mass of a dozen cans of cool drinks	**L**: mg	**M**: t	**G**: kg
12.	the capacity of a bucket	**N**: mL	**R**: L	**T**: kL
13.	the volume of a pizza box	**C**: cm^3	**E**: m^3	**O**: mL
14.	the mass of a pair of gym shoes	**E**: mg	**L**: g	**T**: kg

© EDUCATIONAL DIRECTIONS PUBLICATIONS

WHO DO YOU CALL WHEN YOU WANT HELP WITH YOUR PLUMBING?

Complete each of the following conversions. Look for your answers in the Code Key. Notice the letter next to each answer. Write that letter above the number of the question in the code below.



1. 175 g = _____ kg
2. 82 g = _____ mg
3. 2 850 mg = _____ g
4. 3·75 kg = _____ g
5. 64 kg = _____ g
6. 95 mg = _____ g
7. 5 900 g = _____ kg
8. 0·26 kg = _____ g
9. 690 g = _____ mg
10. 8 g = _____ mg
11. 0·86 kg = _____ g
12. 240 g = _____ kg
13. 5 450 mg = _____ g

CODE KEY

R	– 5·9
M	– 0·008
S	– 82 000
E	– 860
S	– 6·9
U	– 690 000
A	– 0·095
T	– 0·8
R	– 0·175
A	– 5·45
P	– 0·95
F	– 240 000
N	– 64 000
O	– 2·85
L	– 0·082
N	– 3 750
G	– 0·24
D	– 8 000
I	– 260

○	○	○	○	○	○	○	○	○	○	○	○	○	○
6	*10*	*7*	*13*	*8*	*4*	*2*	*9*	*1*	*12*	*11*	*3*	*5*	

© EDUCATIONAL DIRECTIONS PUBLICATIONS

HOW DOES AN ESKIMO BUILD A HOUSE?



Complete each of the following conversions. Look for your answer in the Code Key. Notice the letter next to each answer. Write that letter above the number of the question in the code below.

1. 14·3 t = _ _ _ _ _ _ kg
2. 18·8 kg = _ _ _ _ _ g
3. 22 475 kg = _ _ _ _ t
4. 0·05 kg = _ _ _ _ _ _ g
5. 820 g = _ _ _ _ _ _ _ kg
6. 950 kg = _ _ _ _ _ _ t
7. 775 g = _ _ _ _ _ _ _ kg
8. 73 g = _ _ _ _ _ _ kg
9. 2·81kg = _ _ _ _ _ _ g
10. 0·78 t = _ _ _ _ _ _ kg
11. 475 g = _ _ _ _ _ _ kg
12. 6·2 kg = _ _ _ _ _ _ g
13. 4·865 t = _ _ _ _ _ _ kg
14. 60 kg = _ _ _ _ _ _ _ t
15. 0·65 t = _ _ _ _ _ _ kg
16. 3 950 g = _ _ _ _ _ kg
17. 0·065 t = _ _ _ _ _ kg
18. 3·85 kg = _ _ _ _ _ g
19. 0·5 t = _ _ _ _ _ _ _ kg

CODE KEY

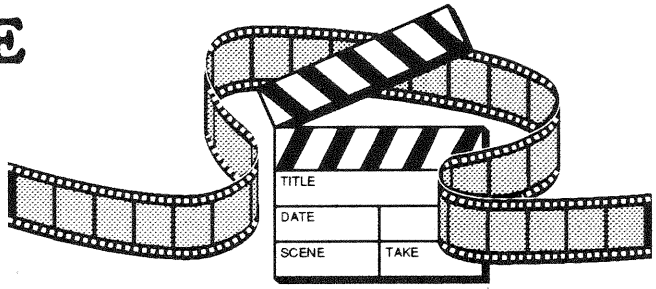
G	650
E	0·95
L	14 300
I	500
E	780
L	0·775
H	65
T	0·82
S	18 800
O	0·06
T	0·073
G	50
A	3·95
I	6 200
L	22·475
0	2 810
T	3 850
R	0·475
O	4 865

12	15	3	9	13	2	19	8	16	7	1

18	14	4	10	5	17	6	11

© EDUCATIONAL DIRECTIONS PUBLICATIONS

WHY DID THE MAN CALL HIMSELF AN EXPERIENCED ACTOR?



Complete each of the following conversions. Look for your answers in the boxes below. Cross out each box containing a correct answer. When you finish, there will be 12 boxes remaining. Write the words in those boxes in order from left to right.

5·5 mg = _____ g

0·072 kg = _____ g

795 g = _____ kg

480 g = _____ kg

3·8 t = _____ kg

6·5 g = _____ mg

13 900 kg = _____ t

800 kg = _____ t

0·62 kg = _____ g

9 mg = _____ g

22·5 mg = _____ g

14·8 kg = _____ g

0·92 t = _____ kg

8 500 kg = _____ t

41·5 t = _____ kg

87·5 g = _____ mg

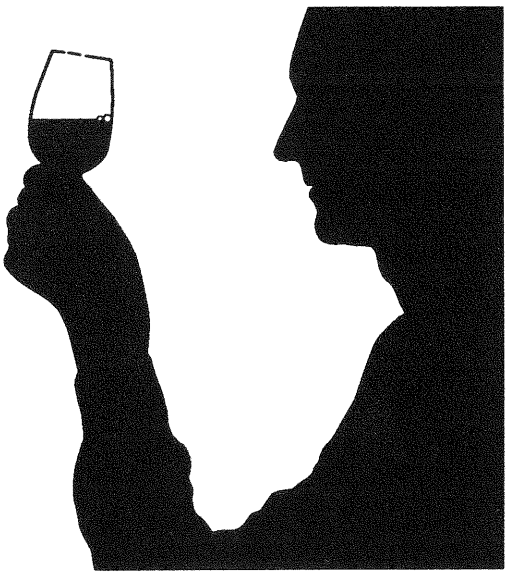
48 g = _____ mg

3 760 g = _____ kg

6 500 ACTOR	0·0875 HE	0·48 STAGE	0·0875 BROKE	0·795 AND	14 800 CAUGHT
148 A	0·8 AND	3 800 PLAY	0·009 SHE	0·0065 LEG	72 MADE
90 AND	0·0055 THEN	8 WAS	3·76 HOSPITAL	920 DID	4 150 IN
92 A	8·5 WRIST	41 500 THEATRE	48 000 WIFE	620 NOT	22 500 CAST
62 FOR	13·9 WATCH	0·0225 PUT	4·8 THREE	87 500 MANY	5 500 MONTHS

© EDUCATIONAL DIRECTIONS PUBLICATIONS

WHAT DO YOU CALL A LEMONADE EXPERT?

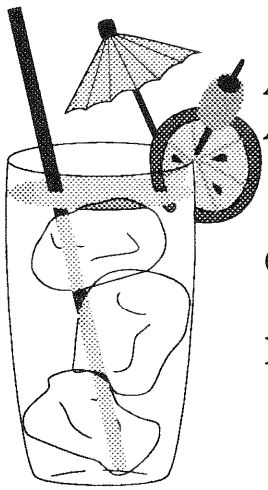


Solve each of the following. Look for your answers in the code below. Write the letter of each question above its corresponding answer.

Z Add 675 kg, 860 kg, 340 kg, 500 kg and 395 kg. Give your answer in tonnes.

F From 4 950 kg subtract 2·2 tonnes. Give your answer in tonnes.

I Multiply 700 g by 8 and give your answer in kg.

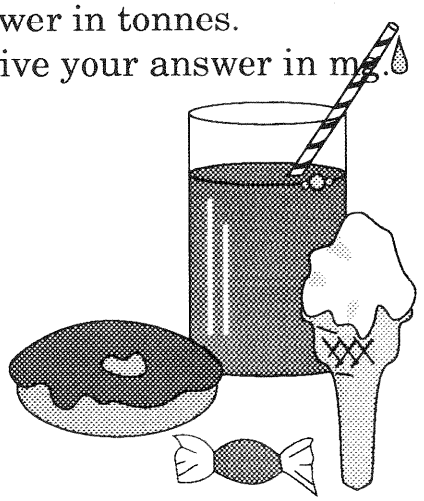


A Divide 8·4 kg by 6. Give your answer in kg.

A Five cakes were cooked for a fete. They weighed 800 g, 750 g, 900 g, 1·3 kg and 1·1 kg. What was the total mass in kilograms?

C What is the difference between 5·9 tonnes and 2 675 kg? Give your answer in tonnes.

N Divide 7·52 g by 5 and give your answer in mg.



I Divide 1·6 kg by 8 and give your answer in kg.

I Multiply 900 g by 5 and give your answer in kg.

Z From the following masses, identify the range:
950 kg, 1·7 t, 860 kg, 2·25 t, 980 kg.
Give your answer in kg.

4·85	**2·75**	**0·2**	**2·77**	**1 390**	**5·6**	**3·225**	**4·5**	**1·4**	**1 504**

© EDUCATIONAL DIRECTIONS PUBLICATIONS

© EDUCATIONAL DIRECTIONS PUBLICATIONS

WHERE DOES TARZAN BUY HIS CLOTHES?

Determine the range of each of the following sets of numbers. Look for your answer in the code below. Write the letter of each question above its range.

J 3, 10, 12, 17, 18
L 5, 8, 15, 25, 31
T 0, 2, 17, 33
A 12, 20, 28, 36
S 18, 37, 54, 62
G 38, 46, 100, 7, 54, 14
U 68, 50, 95, 50, 34, 27, 21
N 72, 72, 16, 33, 81, 47, 19, 56
E 59, 69, 79, 33, 57, 41, 63



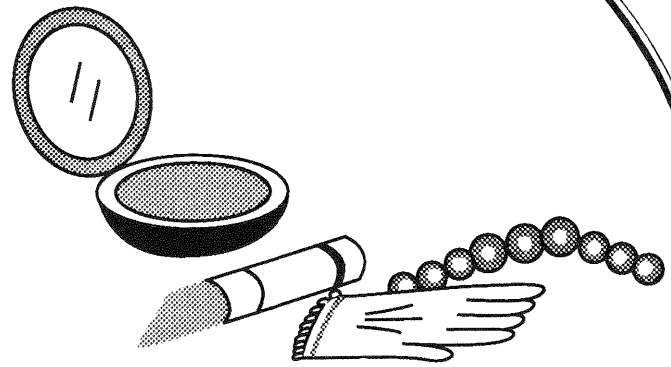
○	○	○	○	○	○	○	○	○	○	○	○	○
24	33	24	15	74	65	93	26	46	44	24	26	46

© EDUCATIONAL DIRECTIONS PUBLICATIONS

WHY DIDN'T THE GIRL USE TOOTHPASTE?

Determine the range of each of the following sets of numbers. Look for your answer in the code below. Write the letter of each question above its range.

L 4, 6, 7, 10, 12, 13
O 4, 16, 22, 28
E 33, 35, 34, 38
S 14, 19, 20, 26, 26, 20
N 41, 46, 50, 52, 52, 68
W 33, 36, 51, 59, 64, 65
F 12, 19, 17, 15, 27, 23, 21
T 87, 89, 91, 92, 88, 88, 94, 94
R 26, 28, 41, 52, 58, 54, 62
H 58, 54, 61, 75, 55, 95, 62, 67

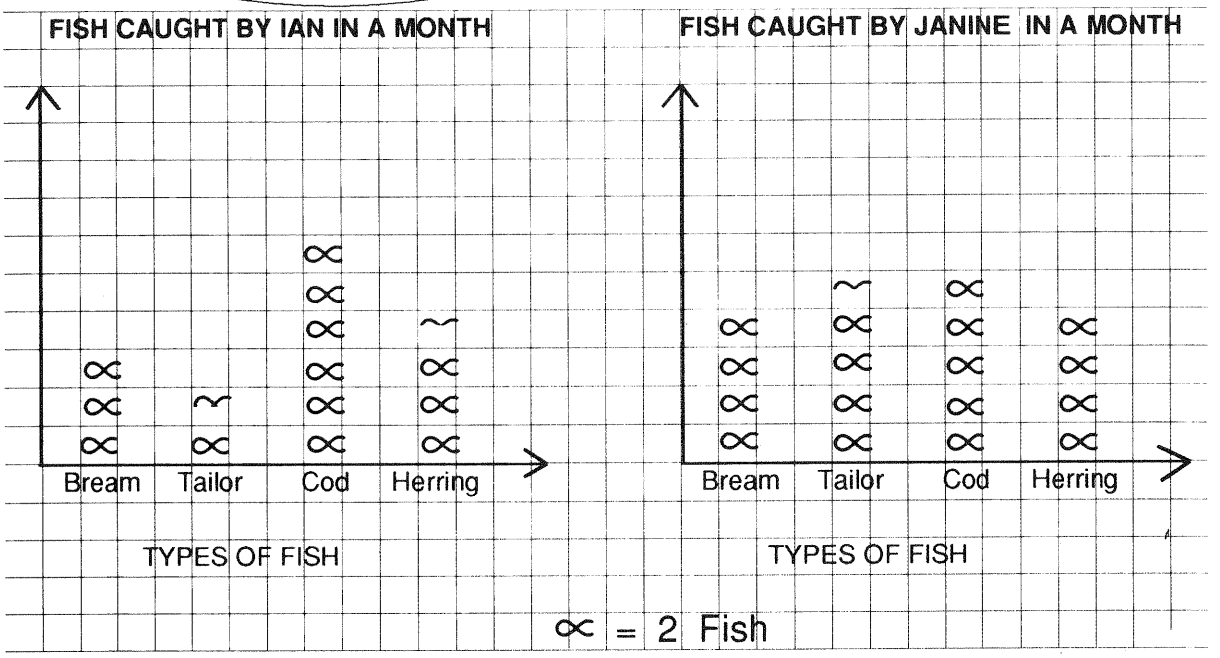


27	24	27	5	24	15	41	5	36	7	5	5	7	41	32	5	36	5

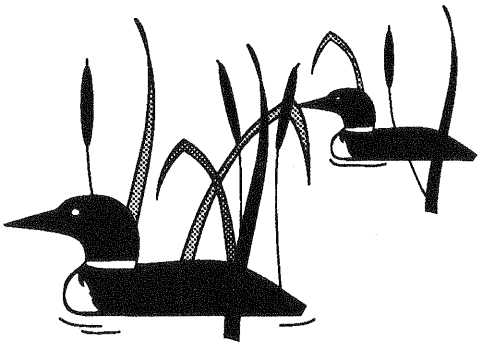
9	24	24	12	5

WHAT IS THE DEFINITION OF A SMART DUCK?

Answer the questions for the following picture graphs. Look for your answers in the grid below. Cross out each box containing a correct answer. When you finish there will be 6 boxes left. Write the letters in those boxes in order from left to right.



1. How many fish did Janine catch?
2. How many more fish than Ian did Janine catch?
3. Overall how many herring did the two people catch that month?
4. What was the total amount of fish caught by Janine and Ian that month?
5. From the graph, which type of fish had the least catch?
6. From the graph, which type of fish had the greatest catch?

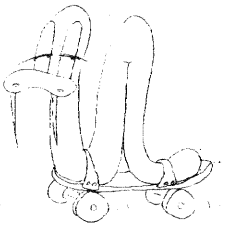


Tailor TR	12 AW	35 EA	Herring IS
15 AC	10 EQ	63 LK	Bream UA
9 CK	Cod ST	8 ER	7 ND

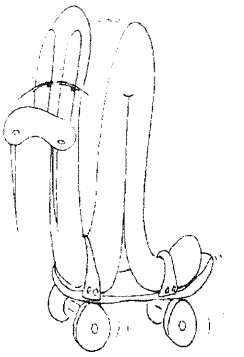
© EDUCATIONAL DIRECTIONS PUBLICATIONS

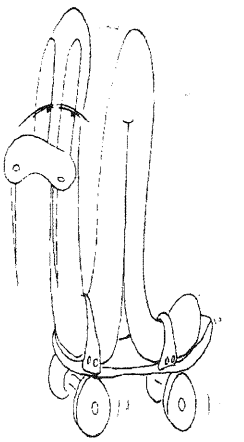
What Do You Get If You Cross A Magician With A Snake?

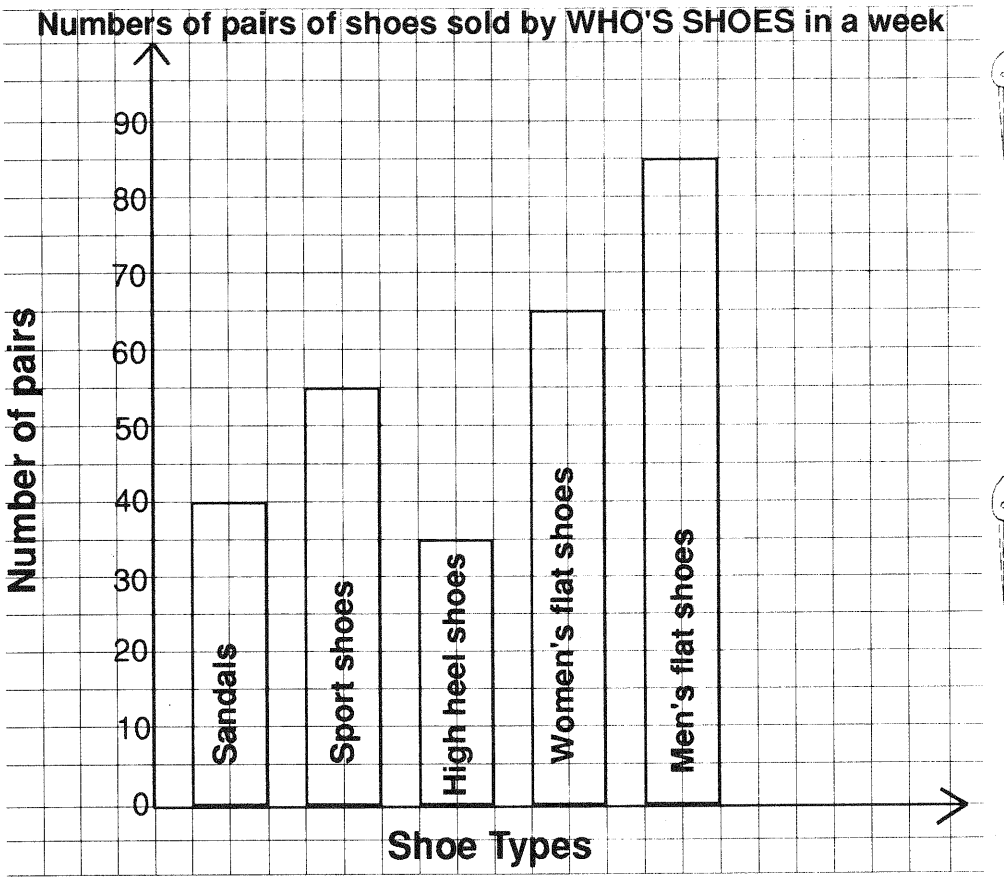
4	6	2	4	1	4	3	5	6	2	4
☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐



Use this graph to answer the following questions. Look for your answers in the Code Key. Notice the letter next to the answer. Write that letter below the number of the question in the puzzle above.







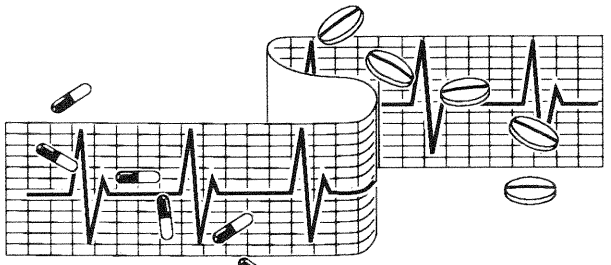
1. How many pairs of sports shoes were sold?
2. How many more pairs of men's flat shoes than women's flat shoes were sold?
3. What was the total number of pairs of shoes sold during the week?
4. What was the most popular type of shoe sold?
5. What was the least popular type of shoe sold?
6. How many types of shoes had sales in excess of 50 pairs?

CODE KEY

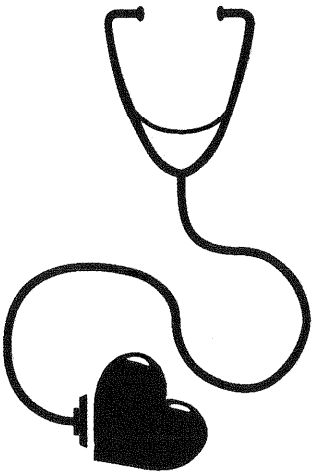
270 – **T**
3 – **B**
265 – **M**
sandals – **L**
high heel shoes – **O**
20 – **R**
4 – **P**
men's flat shoes – **A**
15 – **E**
sport shoes – **W**
women's flat shoes – **G**
280 – **C**
55 – **D**

© EDUCATIONAL DIRECTIONS PUBLICATIONS

WHAT DO YOU GIVE A SICK PIG?



Use these frequency tables to answer the following questions. Look for your answers in the code below. Write the letter of each question above its answer.



Masses of students in class 8G (kg)

Mass (kg)	Tally	Frequency
30 – 39	IIII	4
40 – 49	𝍸 II	
50 – 59	𝍸 𝍸 II	
60 – 69	III	3
70 – 79	I	1

M How many students are in 8G?

N How many students weighed between 40 and 49 kg?

T How many students weighed between 50 and 59 kg?

O How many students weighed more than 49 kg?

Lifetime in hours of a batch of light globes

Lifetime (hrs)	Tally	Frequency
700 – 799	𝍸 III	8
800 – 899	III	3
900 – 999	𝍸 𝍸 IIII	
1 000 – 1 099	𝍸 𝍸 𝍸 𝍸 II	22
1 100 – 1 199	𝍸 I	

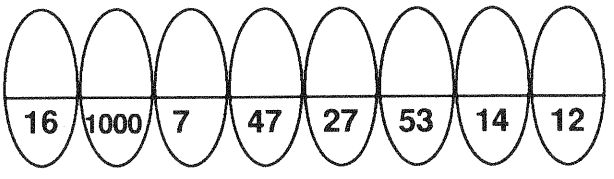
E How many globes are in the batch?

N How many globes had a lifespan between 900 and 999 hours?

K How many globes had a lifespan of less than 1 100 hours?

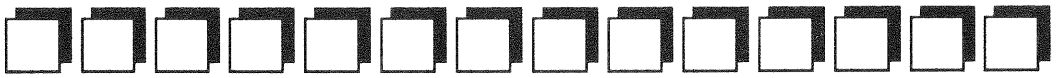


I What might an advertising claim say was the "average" lifespan of these globes?

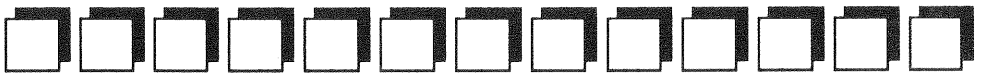


© EDUCATIONAL DIRECTIONS PUBLICATIONS

WHAT SHOULD YOU DO IF YOU SPLIT YOUR SIDES LAUGHING?



39 3 12 3 12 9 166 25 83 163 3 30 50 9



48 178 9 166 9 8 11 166 12 9 11 50 7

Answer the questions for each of the stem-and-leaf plots below. Look for your answers in the code above. Write the letter of the question above its answer.

HEIGHTS OF STUDENTS IN CLASS 8M (cm)

17	0, 1, 2, 3, 6, 8
16	0, 0, 1, 1, 1, 2, 3, 6, 6, 6, 6, 6, 9
15	5, 5, 8, 9
14	8, 9

L How many students are there in the class?
H How many students are taller than 1·6 m but shorter than 1·7 m?
S How tall is the tallest person?
G What is the difference in heights between the tallest and shortest students?
N How many students are taller than 165 cm?
I Five students are the same height. How tall are they?
O When ranked from tallest to shortest, how tall is the student in the middle?

TIMES TAKEN FOR PEOPLE TO COMPLETE A FUN RUN (mins)

4	8, 9
5	0, 0, 1, 2, 3, 8, 8
6	4, 4, 4, 5, 7, 7, 8
7	0, 0, 2, 2, 2, 5, 6, 7
8	1, 1, 3, 3, 3, 3, 3, 3, 5, 8, 9, 9
9	1, 4, 8

R How many people completed the fun run?
T How many people ran in less than one hour?
U How many people took longer than 1·5 hours?
A What was the fastest time?
E What was the difference in times between the fastest and the slowest runners?
Y Six people finished in the same time. What time was that?
M How many people took more than 60 minutes but less than 70 minutes?
C How many people took more than 80 minutes but less than 85 minutes?

© EDUCATIONAL DIRECTIONS PUBLICATIONS

MATH O MAGIC 1·3 SOLUTIONS

1. A CANNIBAL
2. ONLY TRANSISTORS
3. PEN AND OINK
4. NIGHTROGEN
5. BUCK TEETH
6. VANISHING CREAM
7. NEXT PLEASE
8. THE NECKS DOOR NEIGHBOUR
9. LUNATICS
10. SHREDDED TWEET
11. A POOR GOLFER
12. A WIDE ON A TWAIN
13. A CHATTER BOX
14. THE TOOTH FAIRY
15. SHE NEVER GETS THE BEST MAN
16. A HYPNOPOTAMUS
17. A TWO METRE LONG TOOTHBRUSH
18. LIE STILL
19. BREAKFAST IN BED
20. SO HE COULD JINGLE ALL THE WEIGH
21. BECAUSE IT LEAVES THE PATIENT IN STITCHES
22. SHE WAS ALWAYS SLEEPING ON THE JOB
23. A SKELETON CREW
24. A WORDY BIRDY
25. A LADDER IN HER STOCKINGS
26. A DRAIN SURGEON
27. IGLOOS IT ALL TOGETHER
28. HE BROKE A LEG AND WAS IN A CAST FOR THREE MONTHS
29. A FIZZICIAN
30. AT A JUNGLE SALE
31. NONE OF HER TEETH WERE LOOSE
32. A WISE QUACKER
33. ABRA DA COBRA
34. OINKMENT
35. RUN UNTIL YOU GET A STITCH IN THEM

